REPUBLIC POLYTECHNIC

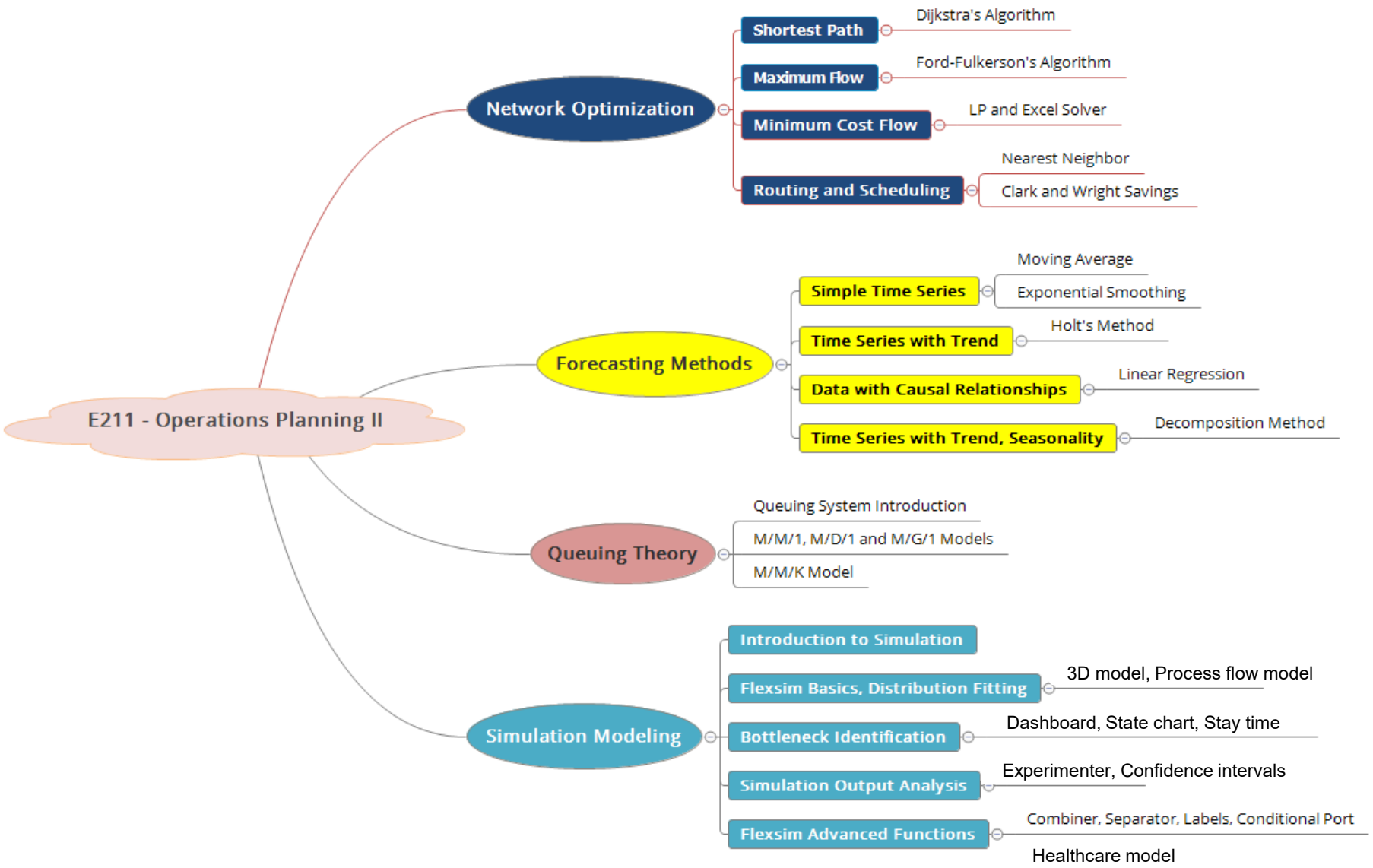
Lesson 9

Food Ordering Process Flow Simulation (I)

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

E211 Topic Tree



Learning Outcomes

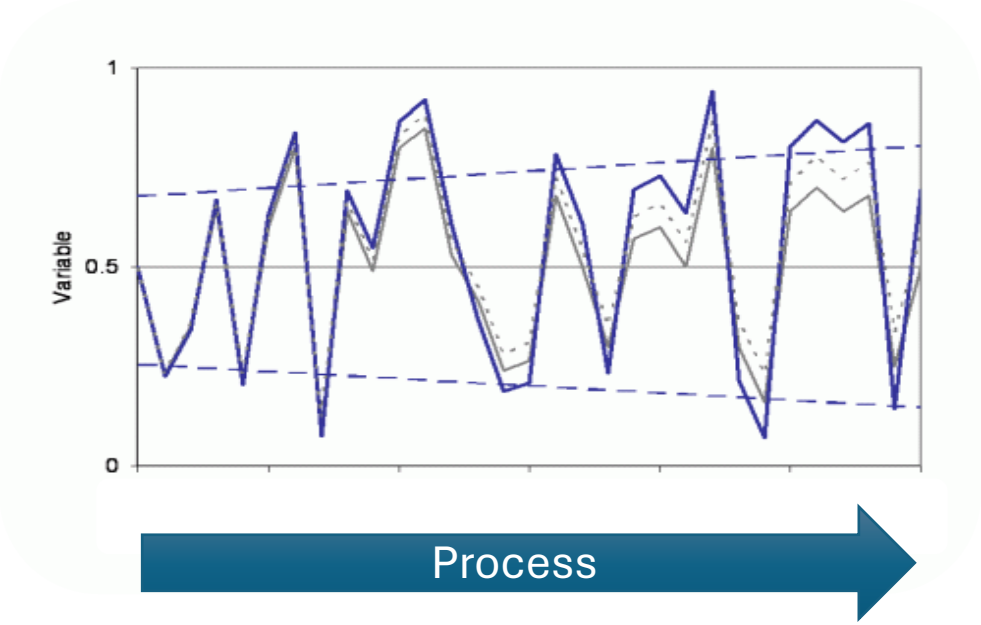
At the end of the lesson, students should be able to:

- Construct a Process Flow model in FlexSim to simulate a multi-stage service system.
- Identify bottleneck in the process through simulation result analysis.

Variability & Waiting Time

- Waiting time is caused by variability in customer arrival or service times, or inadequate service capacity.

- Every service system contains waiting time. Variability propagates through the system and exacerbates congestion downstream of process.



Variability in Systems

- Large variability in system may result in:

  - ✓ Long waiting time and queue length
  - ✓ Increased total time in system
  - ✓ Under-utilized capacity
  - ✓ Decreased service level

- Systems with variability must be buffered by some combination of:

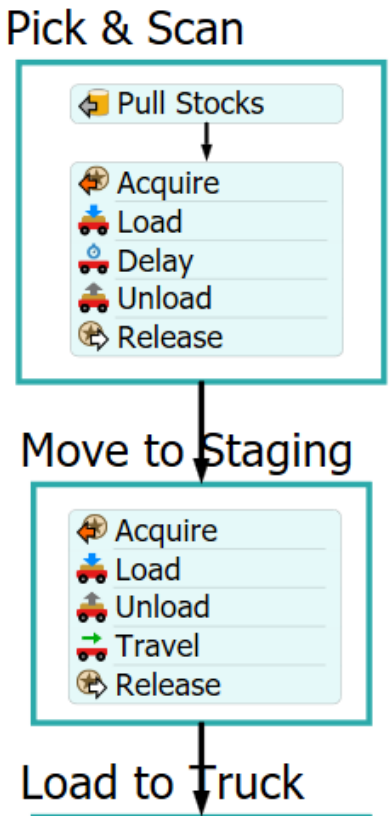
  - ✓ Inventory e.g., some activities carried out in advance
  - ✓ Capacity e.g., extra manpower/ equipment
  - ✓ Time e.g., spreading out arrivals

- Simulation is well-suited to study of systems with large and propagating variability

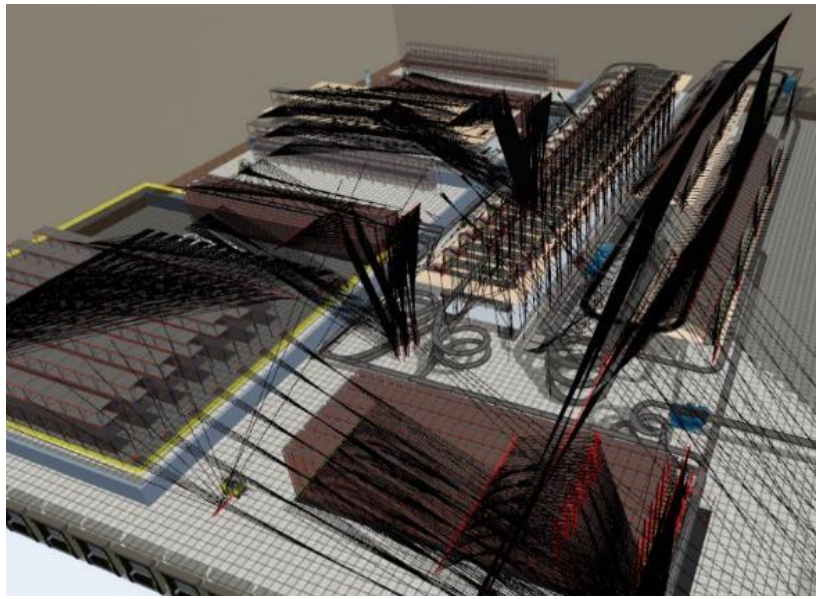
Model Building & Case Scenario

Process Flow Vs 3D Simulation

Process Flow Model



3D Simulation Model



Advantages of Process Flow Model

- **Workflow-Centric Design for Clarity:** Process flow simulation models provide a clear, step-by-step representation of workflows, allowing every process step to be precisely defined and visualized.

- **Quick and Efficient Modeling of Simple Processes:** For straightforward tasks, process flow models can be quickly created and simulated, allowing teams to test and refine processes with minimal effort. This enables rapid prototyping and faster decision-making.

- **Ability to Develop Complex Integrated Models:** Process flow simulations are highly versatile and can incorporate intricate process steps, system logic, and control mechanisms, all within one central model. This makes it possible to scale, modify and manage the simulation of 3D models together with the underlying complex process flows.

Case Scenario

SaladBowl is a food outlet serving delicious and healthy salad bowls. The outlet has a long service counter where customers queue to order while serving staff prepare the salads behind the counter.

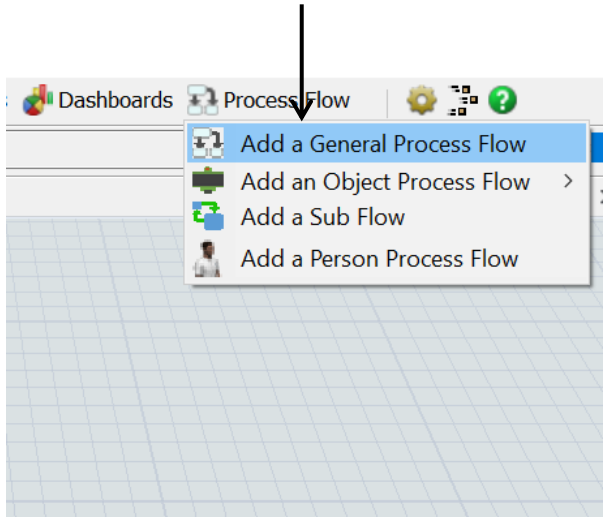
Business at the outlet is generally quite good. And queues are commonly seen during the peak periods. Generally, it was observed that customers will not join the queue to order food if there are already 8 people in the queue. SaladBowl is thinking of ways to reduce the number of balking customers and minimize potential loss of revenue.

Your task is to build a simulation model of the ordering process. The model would be used study ways to improve the performance measures and reduce the revenue loss.

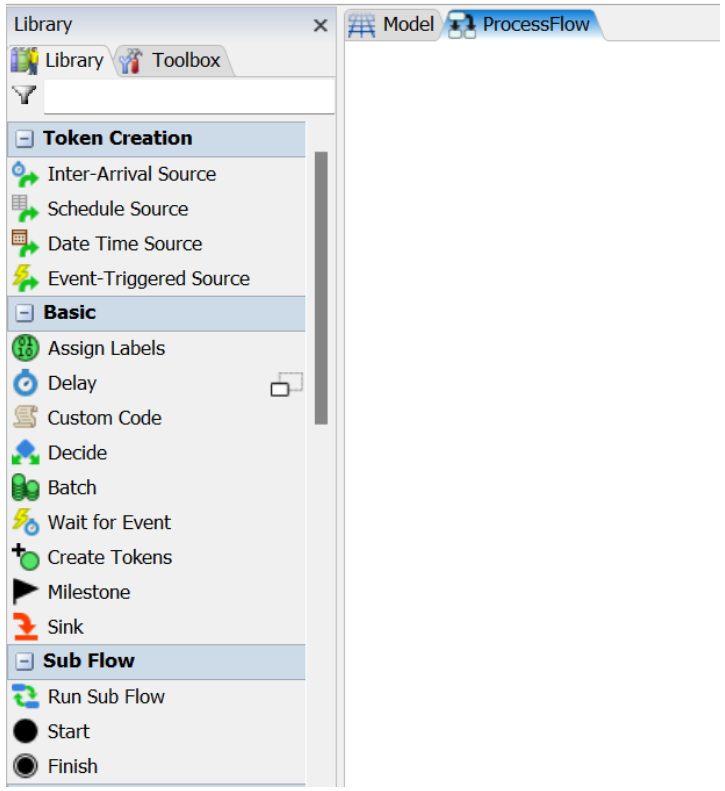
Refer to 'AY 2025 Sem 2 E211 L9 Scenario' for more information.

Starting a Process Flow Model in FlexSim

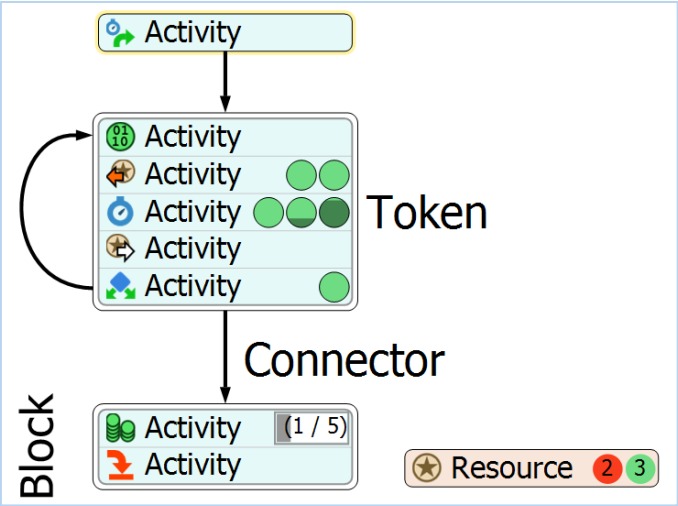
Click on Process Flow and Add a General Process Flow

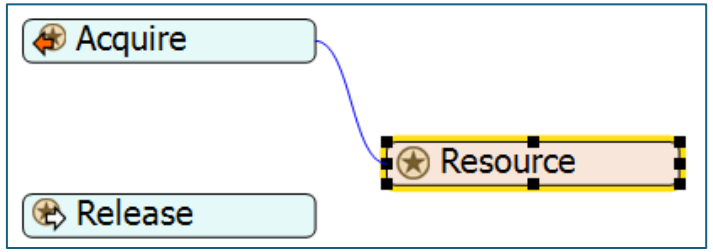


Notice the library changes to process flow objects (activities)



Process Flow Terminologies





Token

- Tokens are the entities that travel through the activities of a process flow. They function similarly to flowitems in a 3D model, but can be anything including information

Activity

- Process boxes that tokens flow through. Includes Assign Labels, Delay, Batch, Decide, etc

Connector

Block

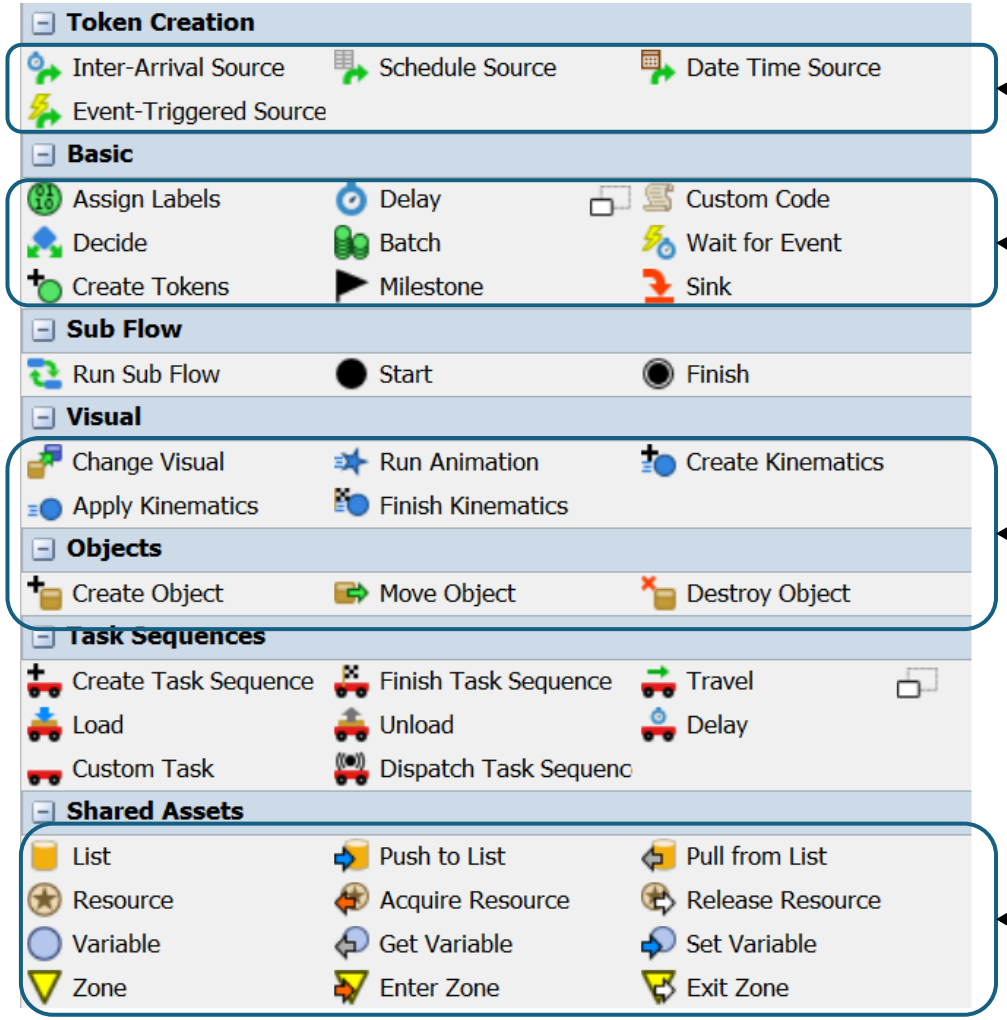
- Show a sequential relationship of activities
- Always enter a block at the head, exit through the foot

Shared Asset

- Some activities may require resources
- Acquire resource before activity
- Release resource on completion of activity

REPUBLIC POLYTECHNIC

Process Flow Activity Library



Token Creation:
To generate tokens in the model

Basic:
Common activities in Process Flow

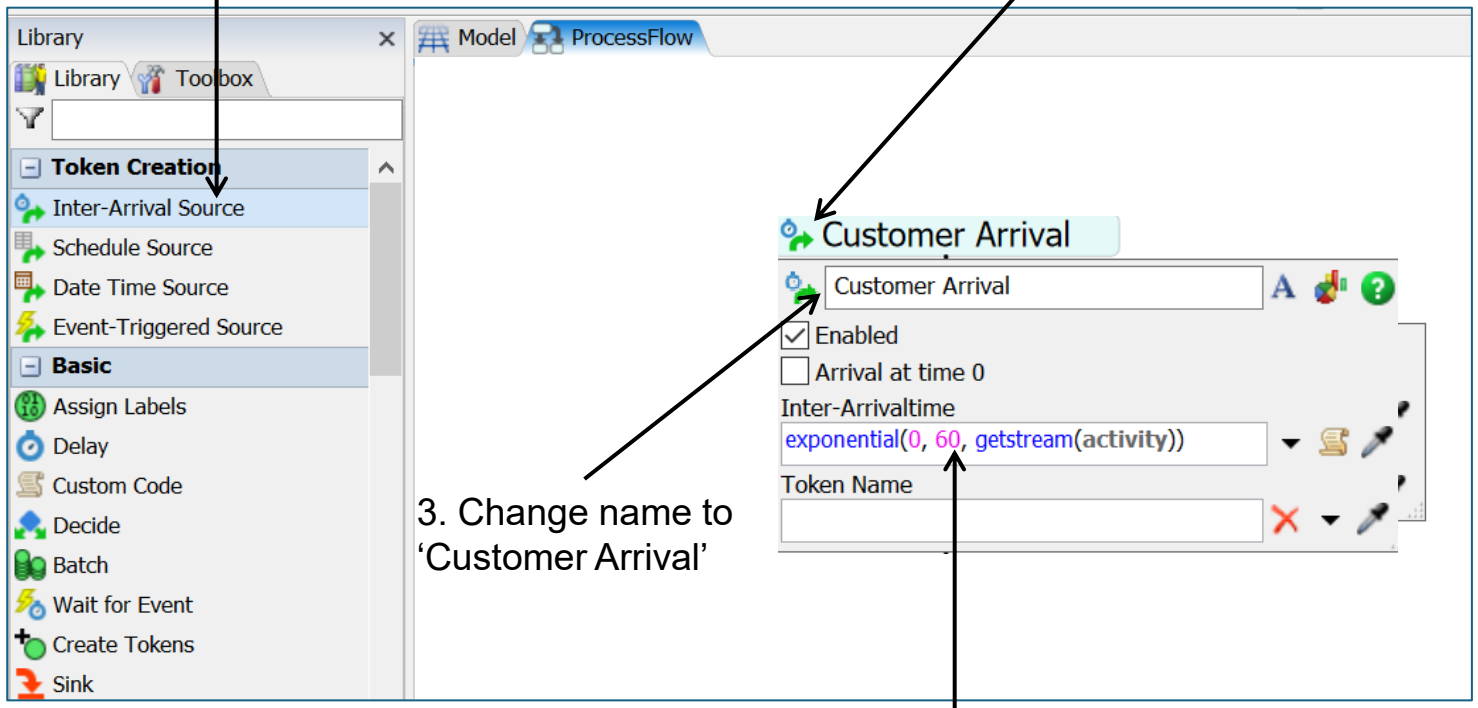
Visual/ Objects:
Used to control 3D objects & items

Shared Assets:
For modeling shared resources

Add Source for Arrivals

1. Click and drag into model window

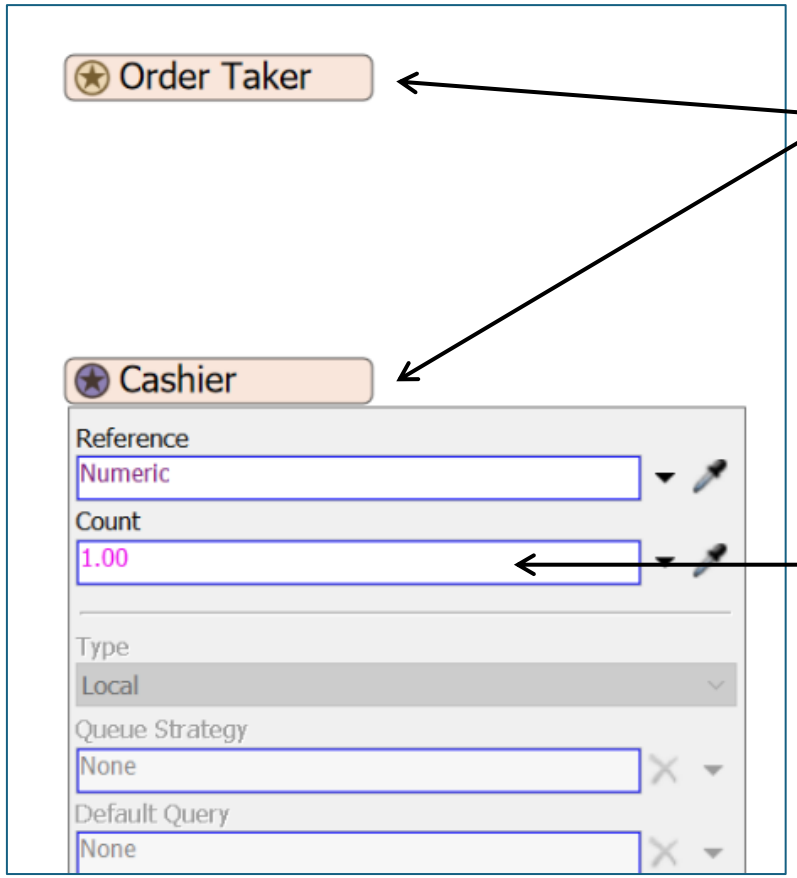
2. Click on icon to access the activity properties (or access from properties window on the right panel)



3. Change name to 'Customer Arrival'

4. Change the 2nd parameter to '60'. What does the value represent?

Add Resources (Staff)



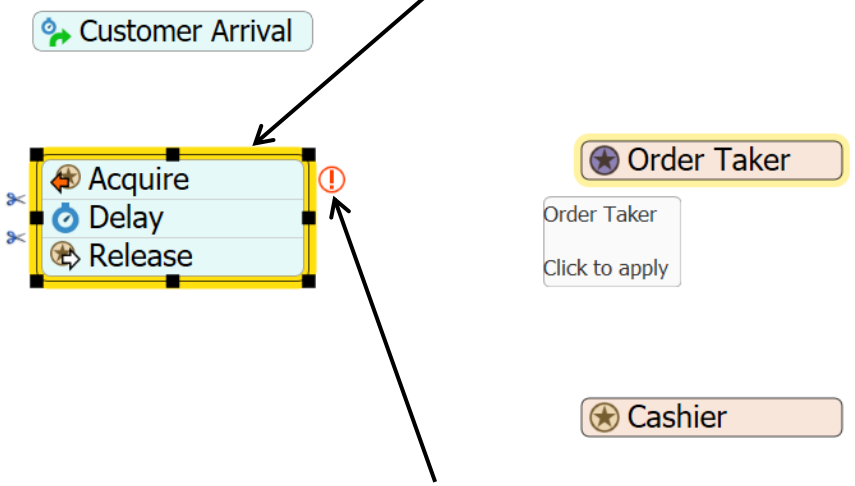
Drag 2 Resources (from Shared Assets) into model.

Rename these as Order Taker and Cashier respectively.

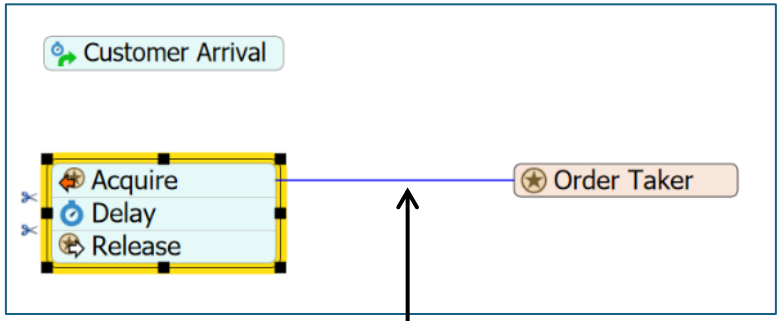
Set Count = 1 for each resource (number of resource available)

Link Resource to Acquire

1. Create Acquire, Delay and Release. Stack them into a Block.



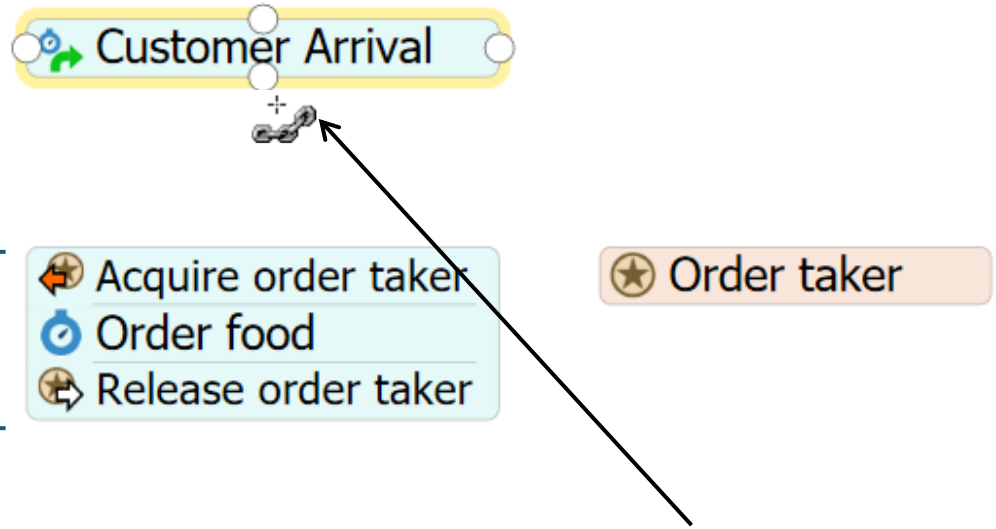
2. Exclamation mark means need to link up Acquire with a Resource. Click on '!' and then click the Order Taker resource.



3. Once linked up, you will see a blue line connection when you click on Acquire or Order Taker.

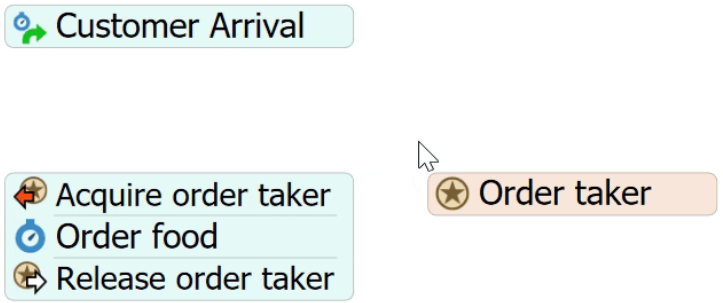
*Note: Release activity is required to free up the resource after use

Connect Activities



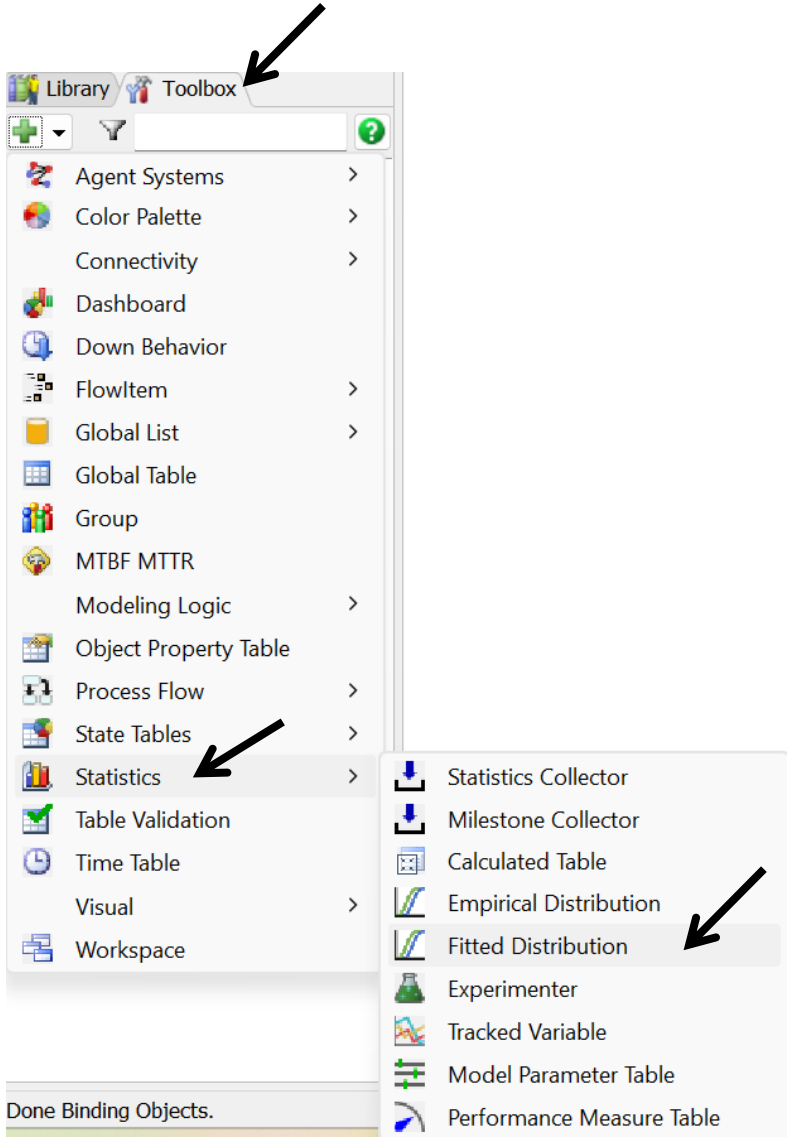
Rename accordingly

To connect the activities, move the mouse cursor close to the box. When the cursor changes to a chain symbol, click and drag to the other box to form the connection.

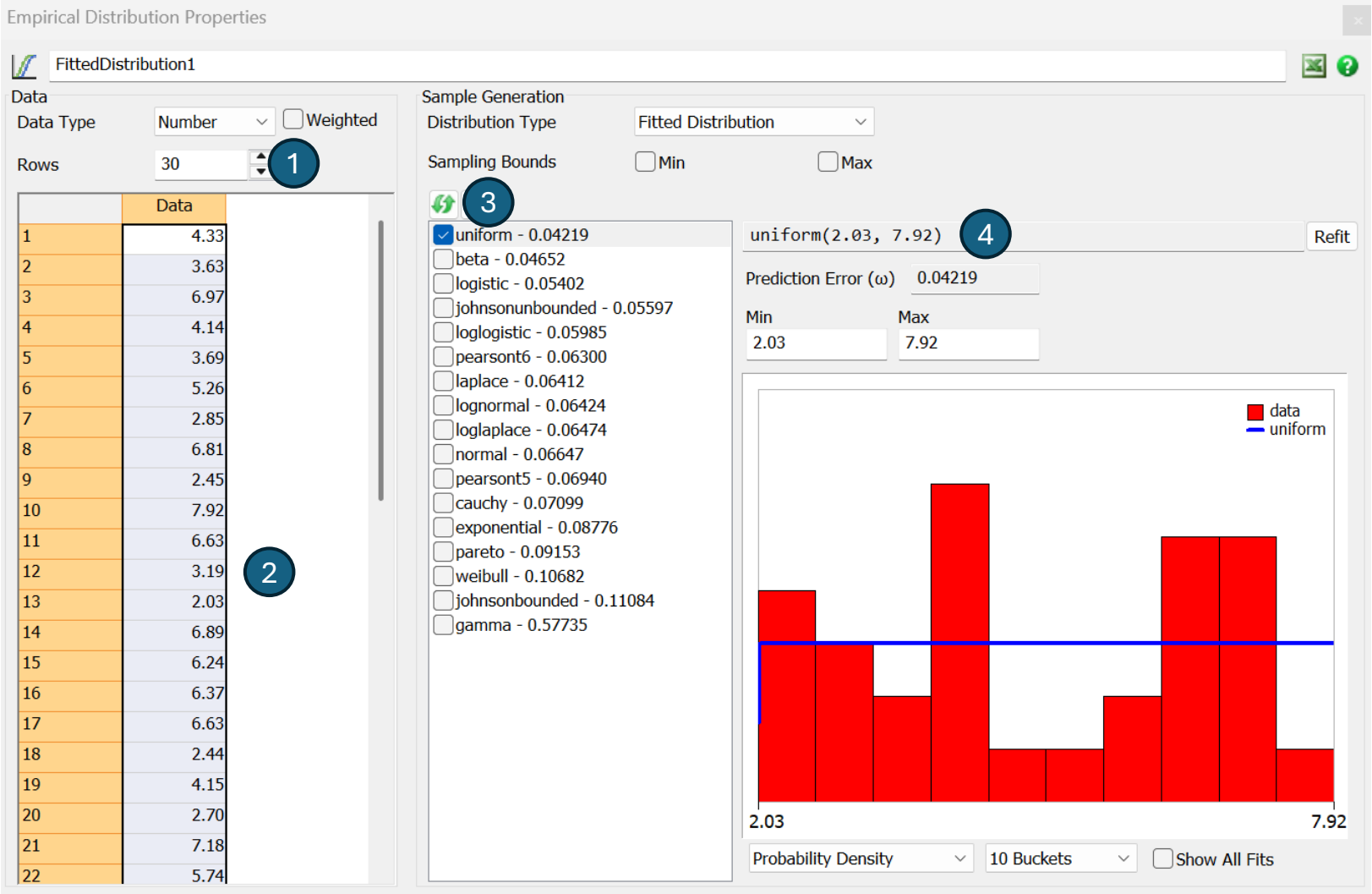


Fitting a Distribution to Data Set

- In simulation, data are often gathered for random processes. E.g. activities with random timings
- A statistical tool can be used to determine a suitable probability distribution to fit a set of data.
- In FlexSim, you may use the 'Fitted Distribution' in Toolbox to generate a probability distribution in order to fit a set of input data.



Fitting a Distribution to Data Set



1. Key in number of data points
2. Enter the data (copy from Excel*)
3. Click ‘refit’ icon. Select the appropriate distribution. The distributions are sorted in ascending order of prediction error.
4. Use the parameters of the distribution in FlexSim.

**Use the Excel file ‘Data for distribution fitting’ to try out this feature.*

When to Use Triangular Distribution

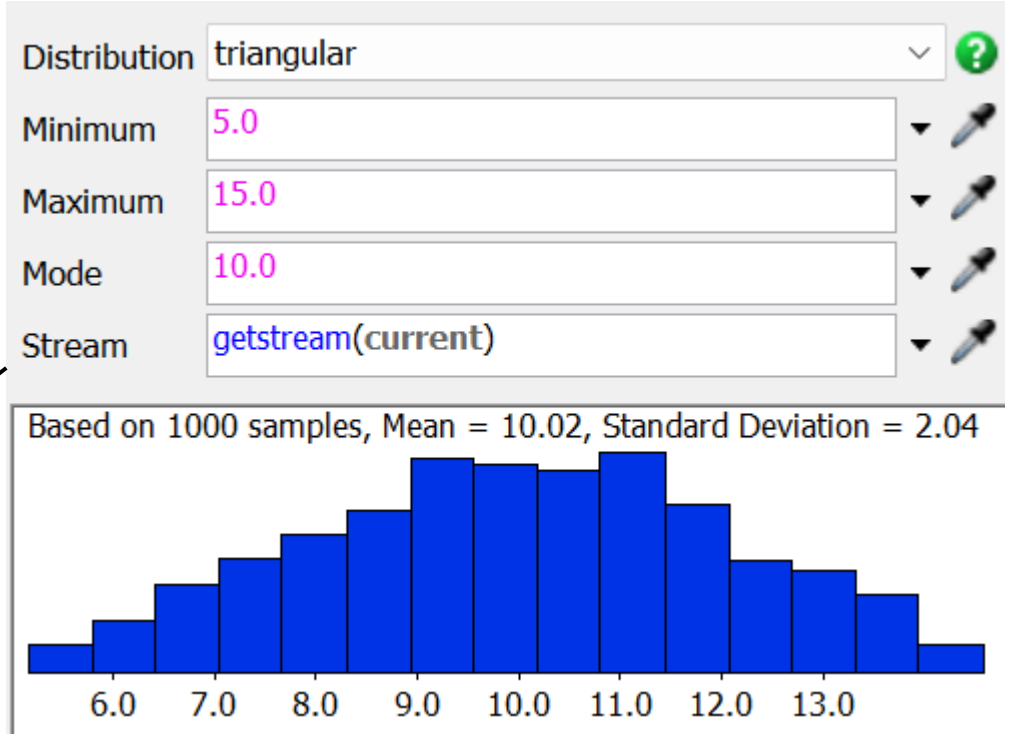
- Obtaining time data may sometimes be difficult. The process might be new, or it is too time consuming to collect a set of representative data.
- In such cases, it is recommended to model the data using a triangular distribution. Its parameters – minimum, maximum and most likely (mode) – can be easily estimated by subject matter experts.

Process Time

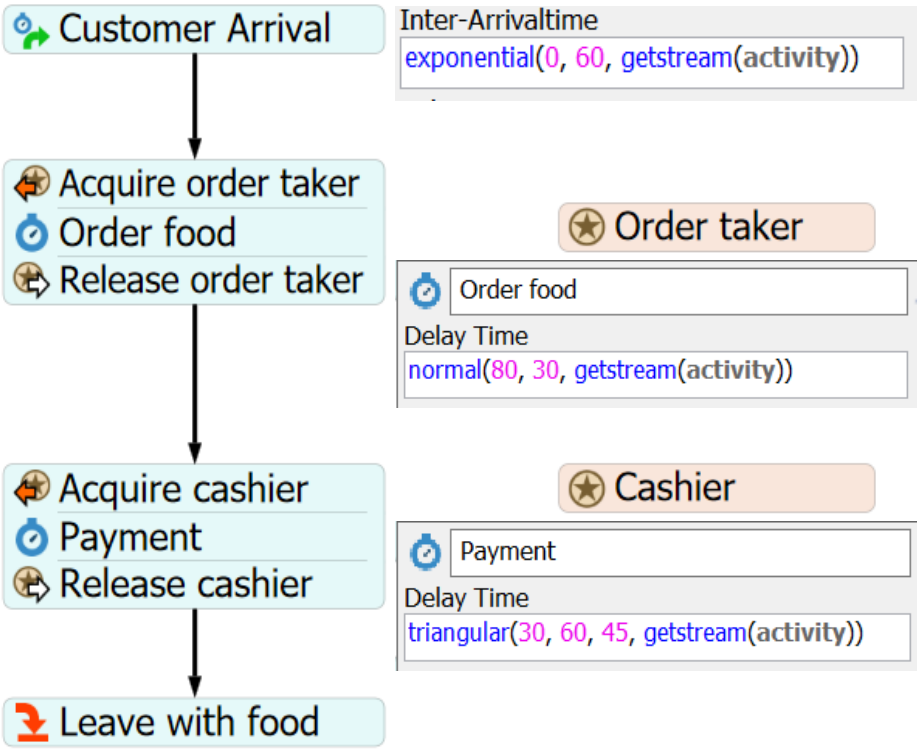
triangular(5.0, 15.0, 10.0, getstream(current)) s

getstream(current) is a function in FlexSim used to generate random number stream

Parameters of Triangular Distribution are Min, Max and Mode



Build and Run Simulation

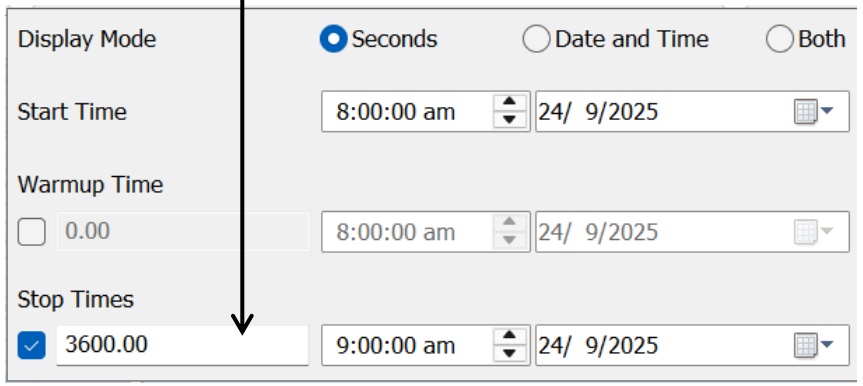


1. Create the rest of the model for and rename them accordingly. Make sure Acquire cashier is linked to Cashier.

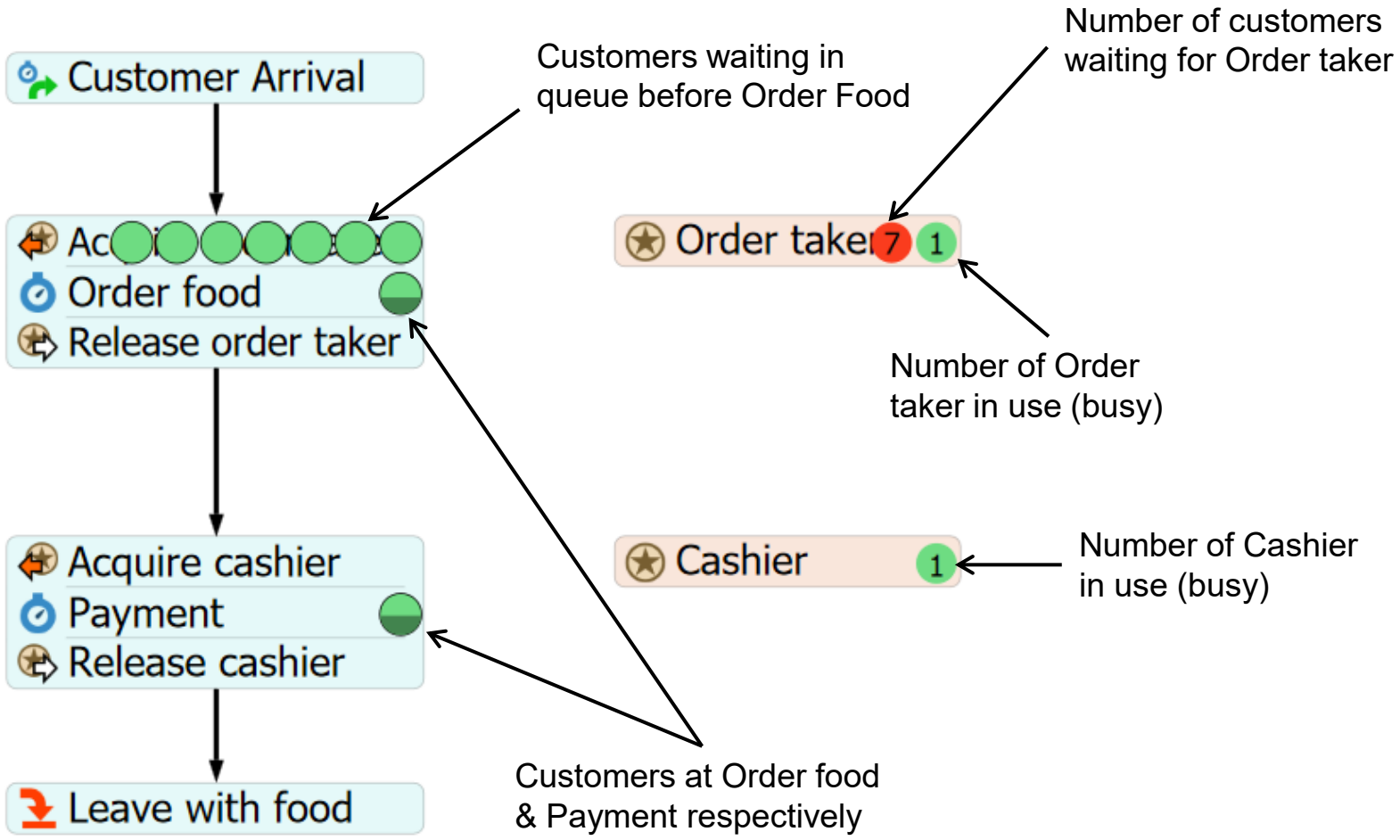
2. Connect the activities and key in the Delay Time values for Order food and for Payment

3. Set Stop Time to 3600 seconds

4. Run the simulation!



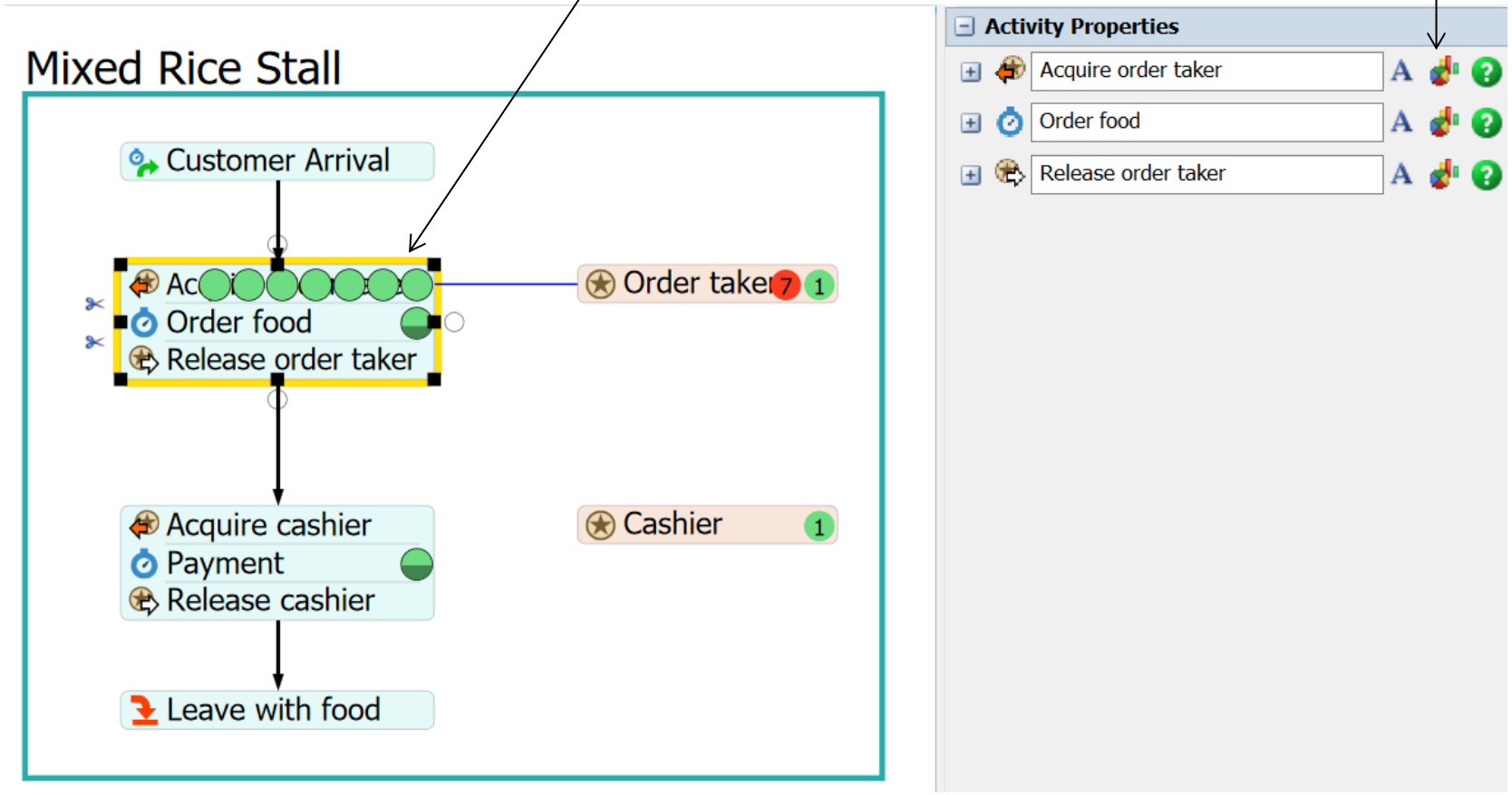
How to Interpret Process Flow Simulation



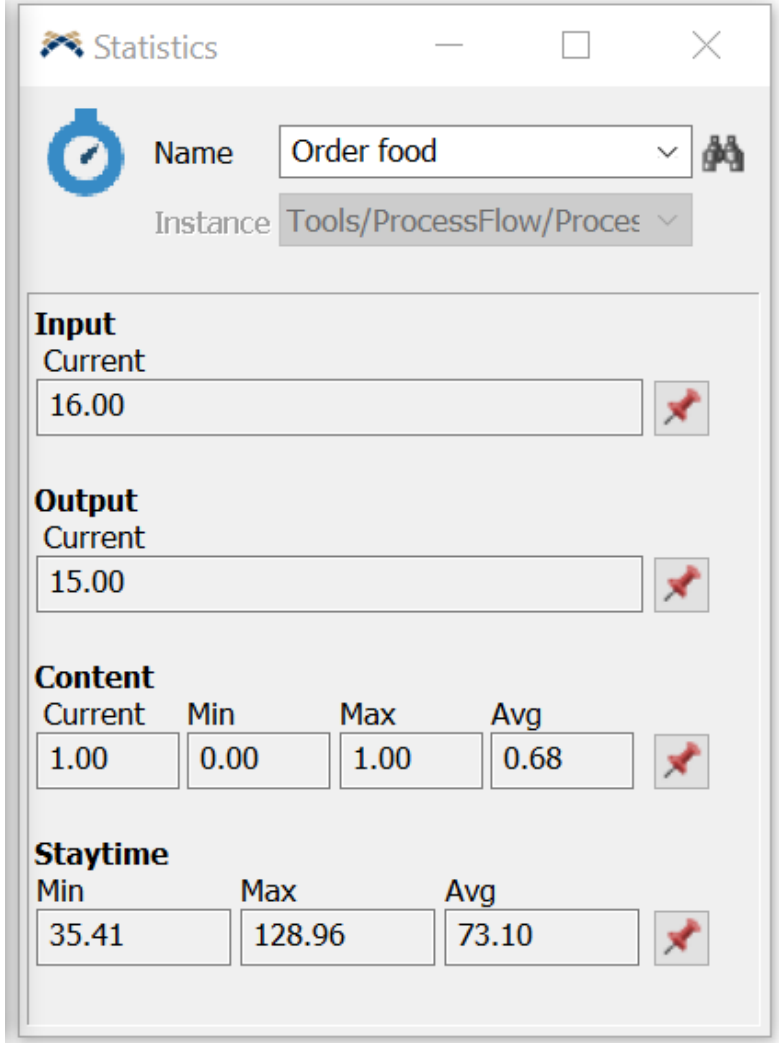
Obtain Simulation Statistics

Click on box to bring up properties window.

Click on stats icon to display activity stats



Statistics for ‘Order Food’ (Delay)



In **Delay** activity, the token will ‘pause’ for a period according to the time delay value set.

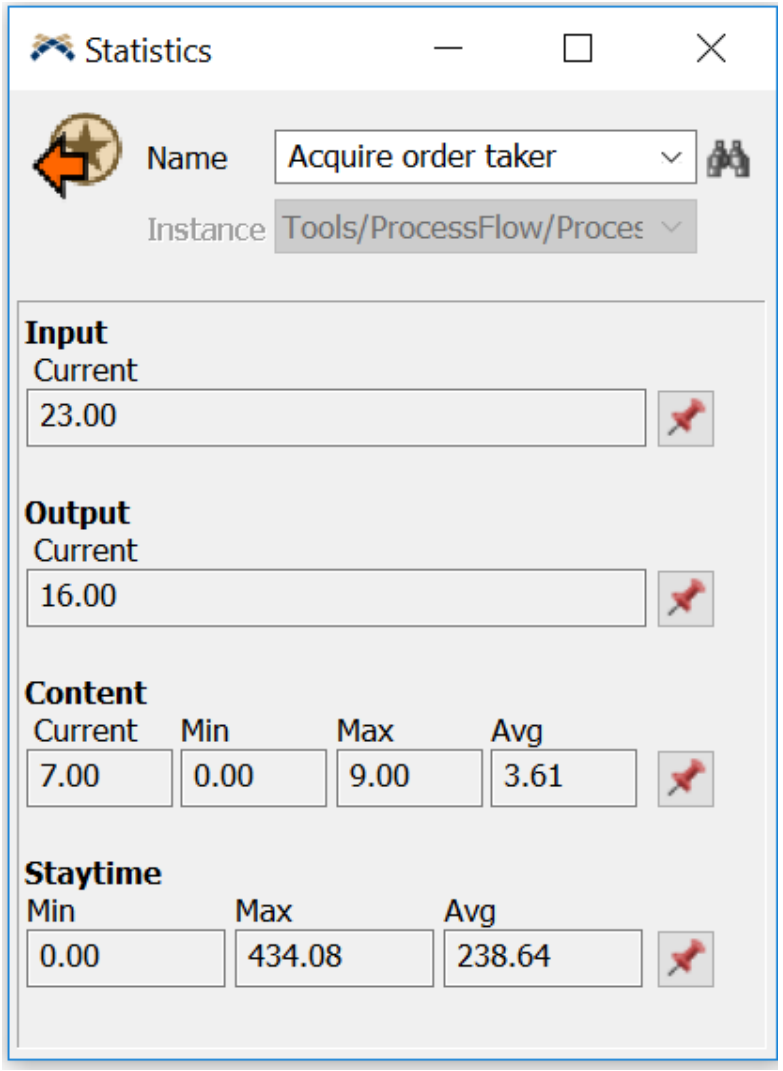
Input: Number entered

Output: Number exited

Content – Number residing in the activity

Staytime – How long did the token stay in the Delay activity

Acquire Resource Statistics



In **Acquire Resource,** the token acts to acquire the linked resource before it can move to the next activity.

Input: Number of tokens requesting for the Resource

Output: Number of token requests fulfilled

Content – Number of token requests at any time

Staytime – How long did the token wait before a request is fulfilled (resource acquired by token)

(Waiting time - Request staytime)
(Queue length - Request Content)
(Order taker performs order taking - Staytime)
Average utilization of this resource = Average content/Max Resource

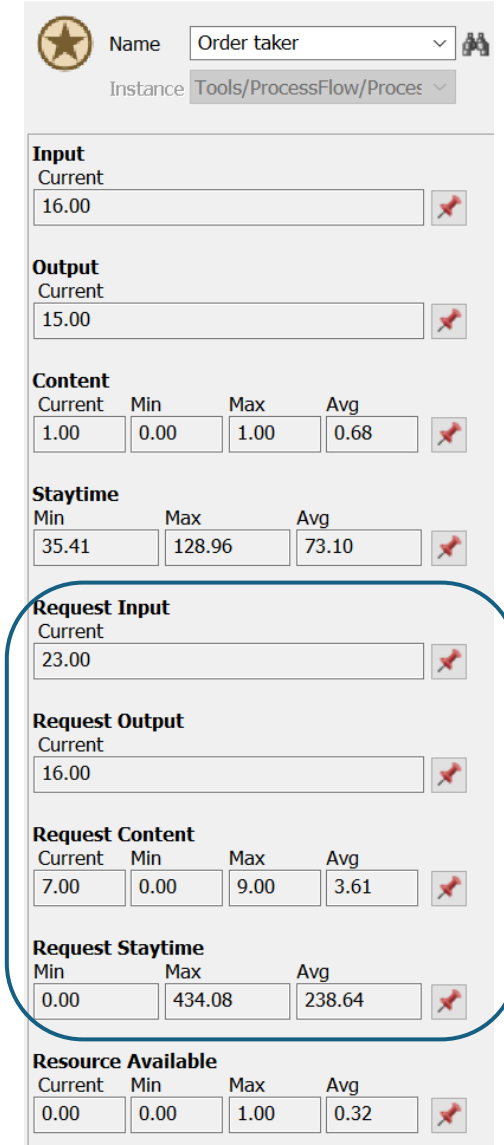
Maximum waiting time (OT): 434.08
Average waiting time (OT): 238.64

Max no of people waiting in queue: 9
Average queue length: 3.61

Average time order taker performs order taking: 73.1

Average utilization of this resource: 0.68
[Average content/Max Resource = 0.68/1 = 0.68]

Resource Statistics



Input: Number processed by Resource (in-process + completed)

Output: Number processed by Resource (completed)

Content: Number of Resource in use at any time.

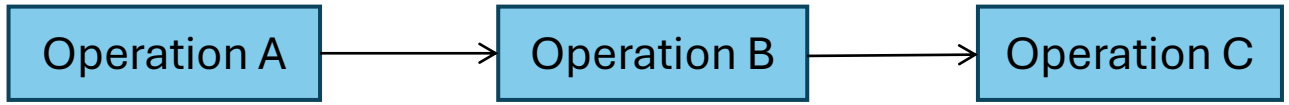
Staytime: From the time the Resource is acquired to the time it is released (duration that the resource is in use).

'Request …': Same statistics as the associated Acquire Resource.

Resource Available: Number of Resource available at any time.

Identifying Bottleneck

- In a process with sequential operations,



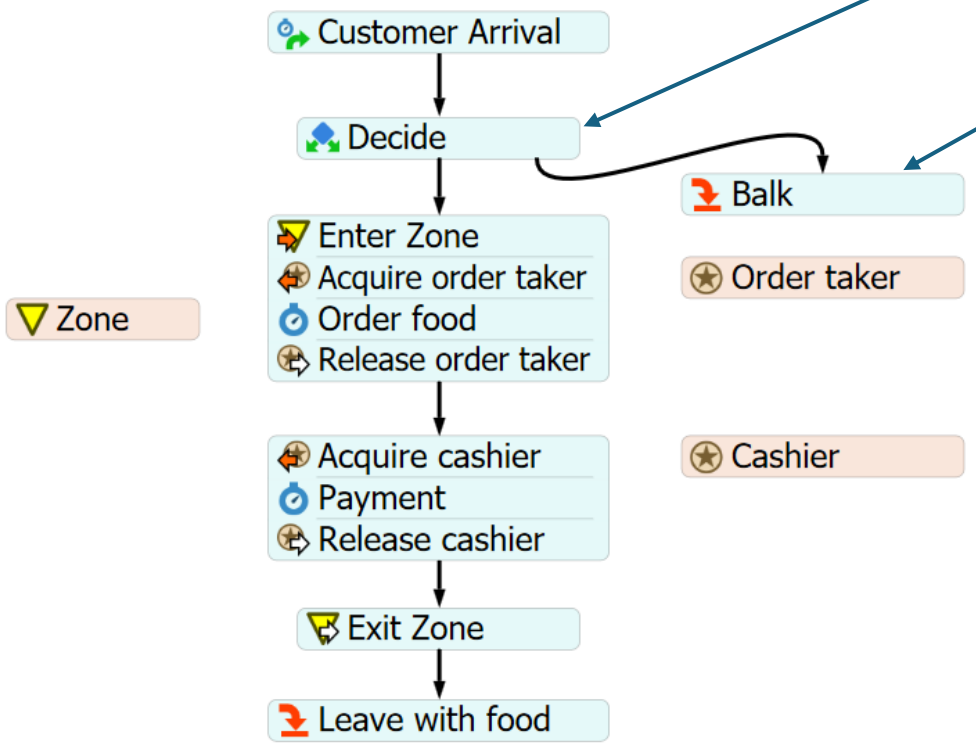
- Bottleneck operation is the one with the longest waiting time, longest queue length and highest utilization.

- The bottleneck can be alleviated by increasing amount of resource to increase capacity or by reducing the process time of the bottleneck operation.

- Improving the performance of non-bottleneck operations will have no impact on system performance.

Model Balk Behaviour (I)

Video:
https://youtu.be/hyb5Ifpr9Ak

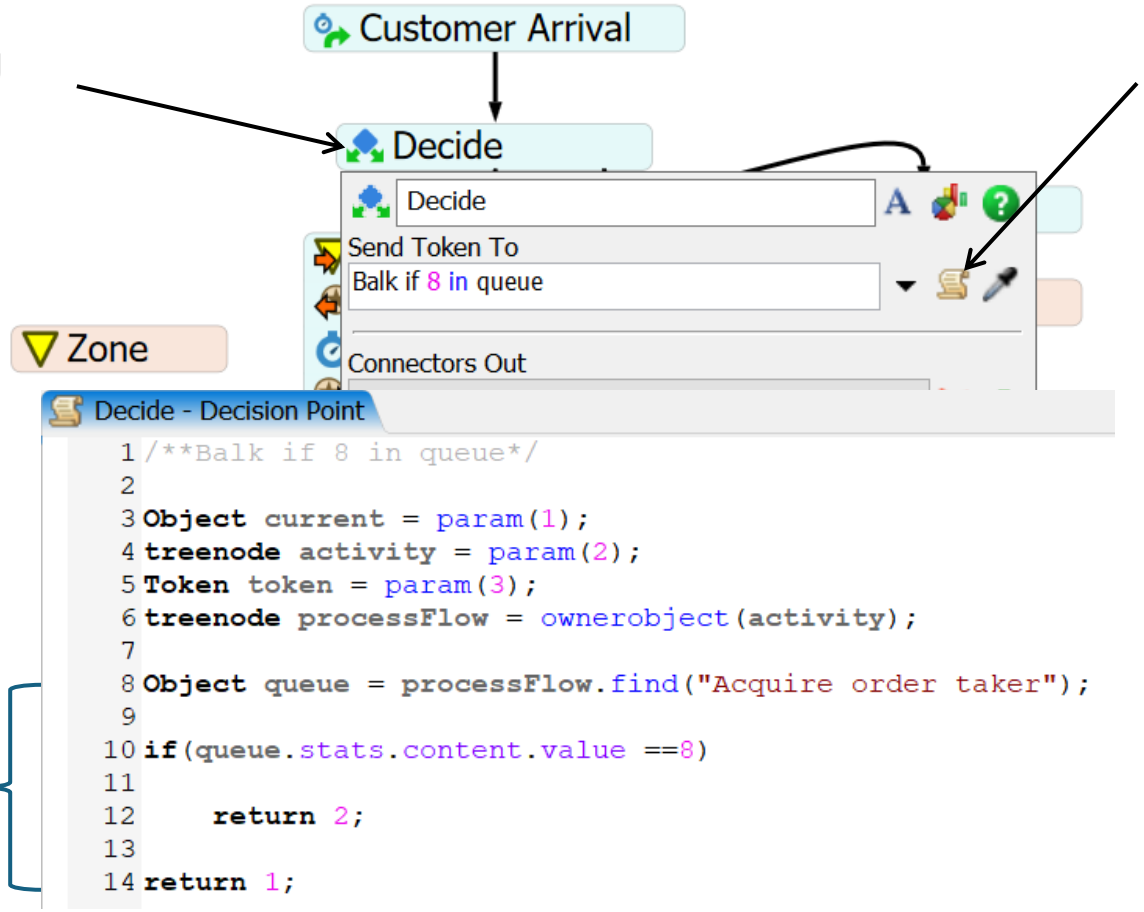


1. Insert a Decide activity from Basic as shown
2. Insert a Sink and rename as Balk
3. Connect Decide to Balk. Connection number should be #2

Model Balk Behaviour (II)

1. Click icon to bring up the properties

2. Click for Code Editor



3. Add codes (Lines 8 to 14)

What the codes mean:
If the number of tokens in Acquire order taker = 8, send token to Balk (Connector #2)

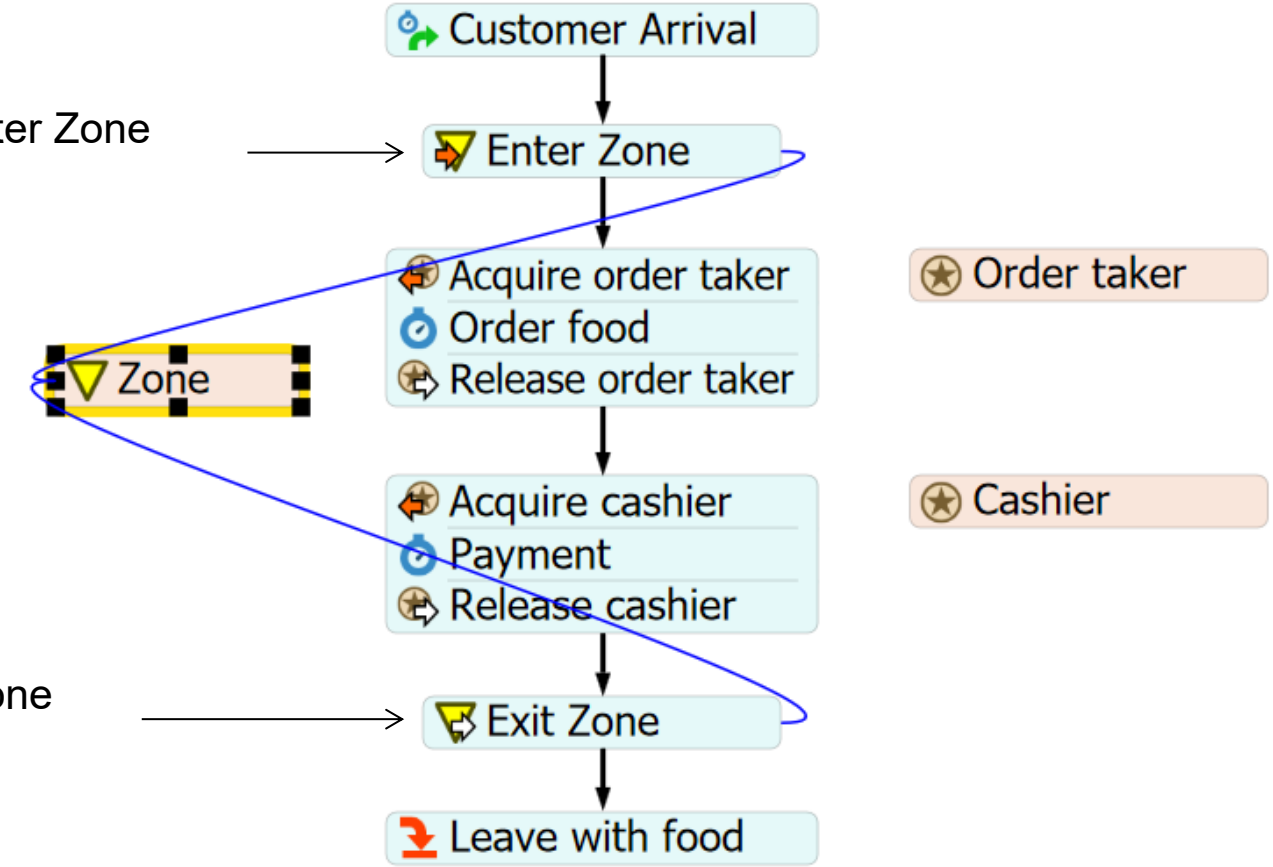
Define Zone

Purpose of zone is to enable statistics to be collected in the Zone (between Enter Zone and Exit Zone).

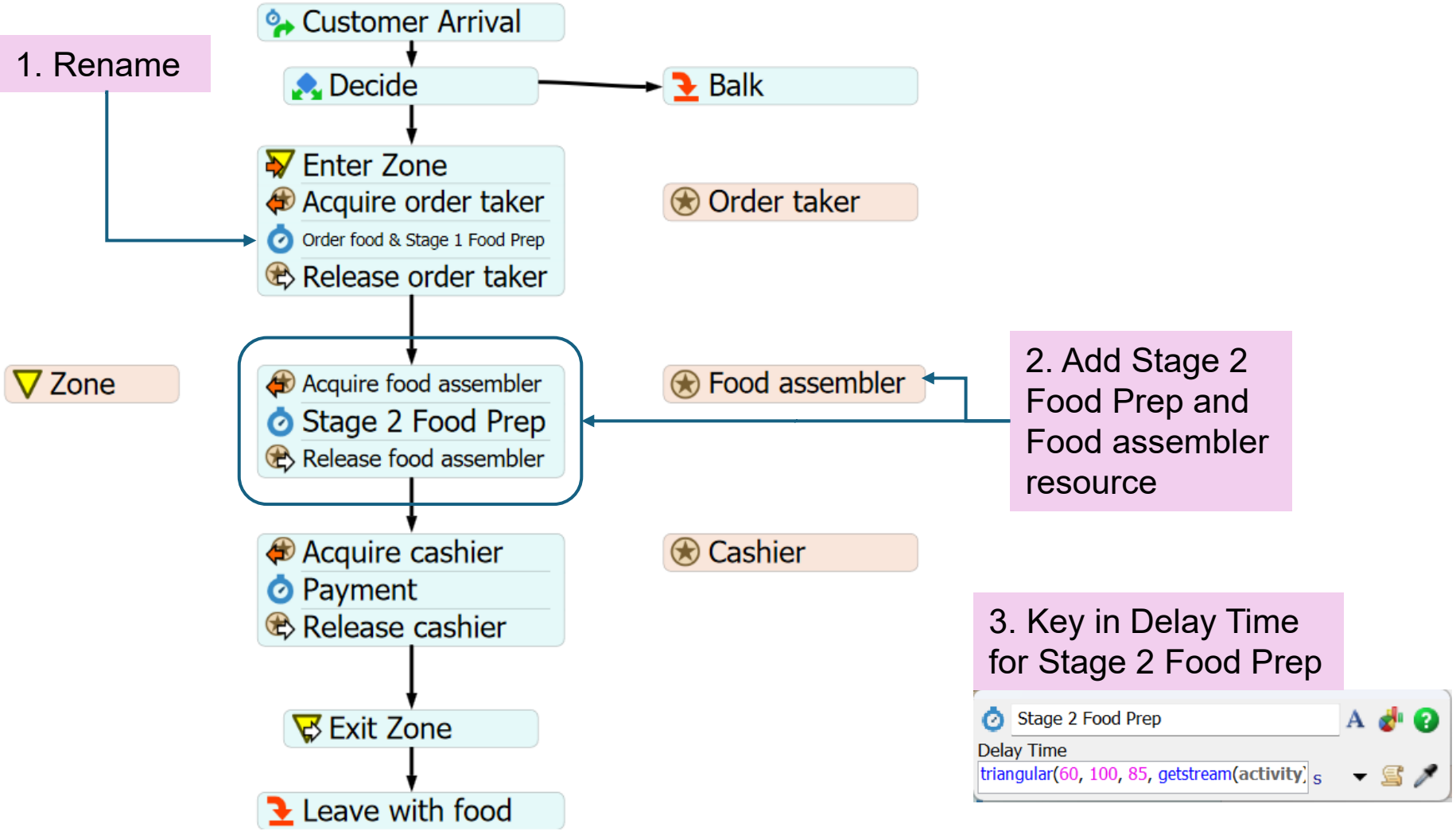
Add Enter Zone here

Link Enter Zone and Exit Zone to Zone

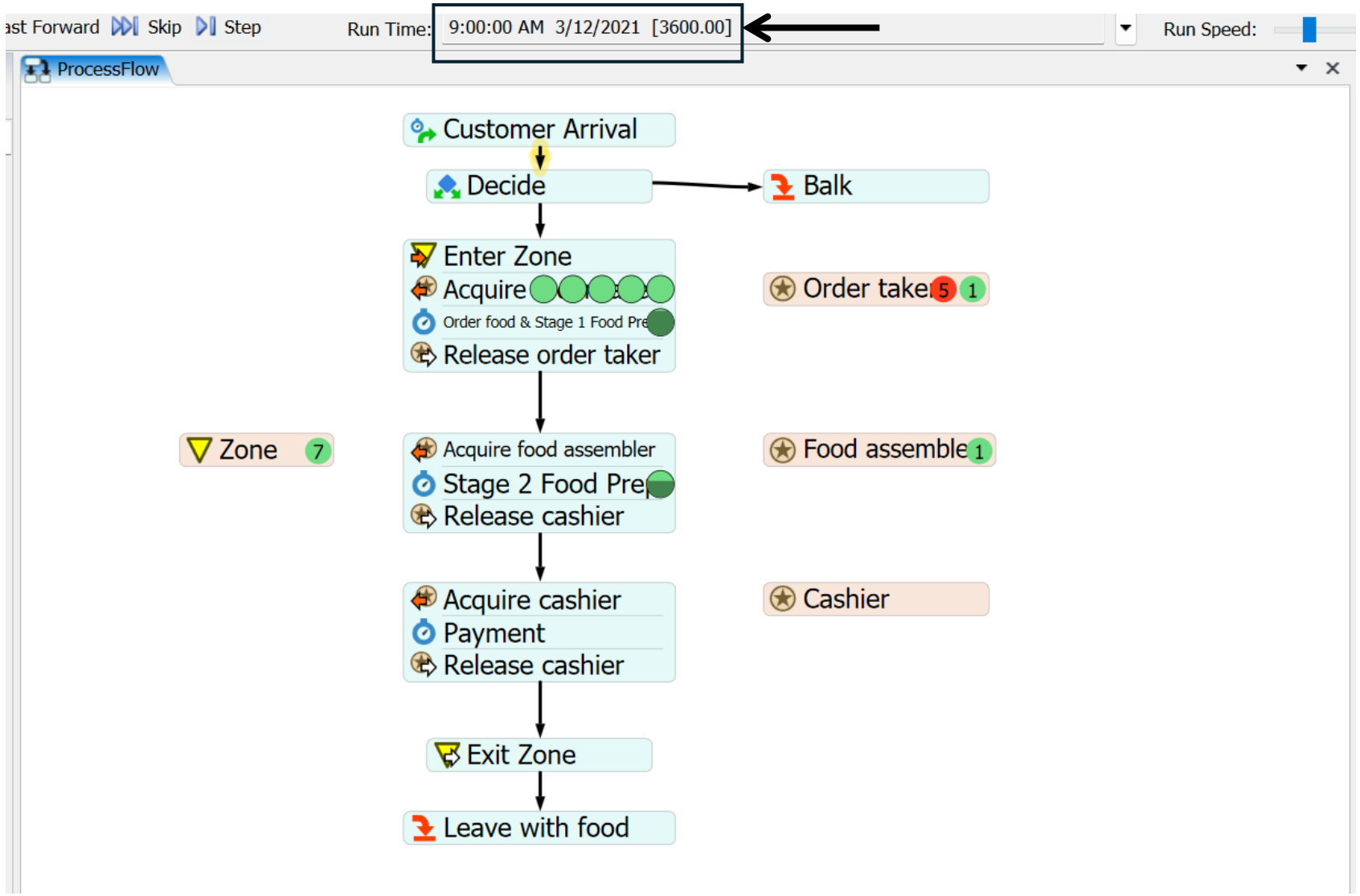
Add Exit Zone here



Model the Rest of the Process



End State of Simulation



Quiz

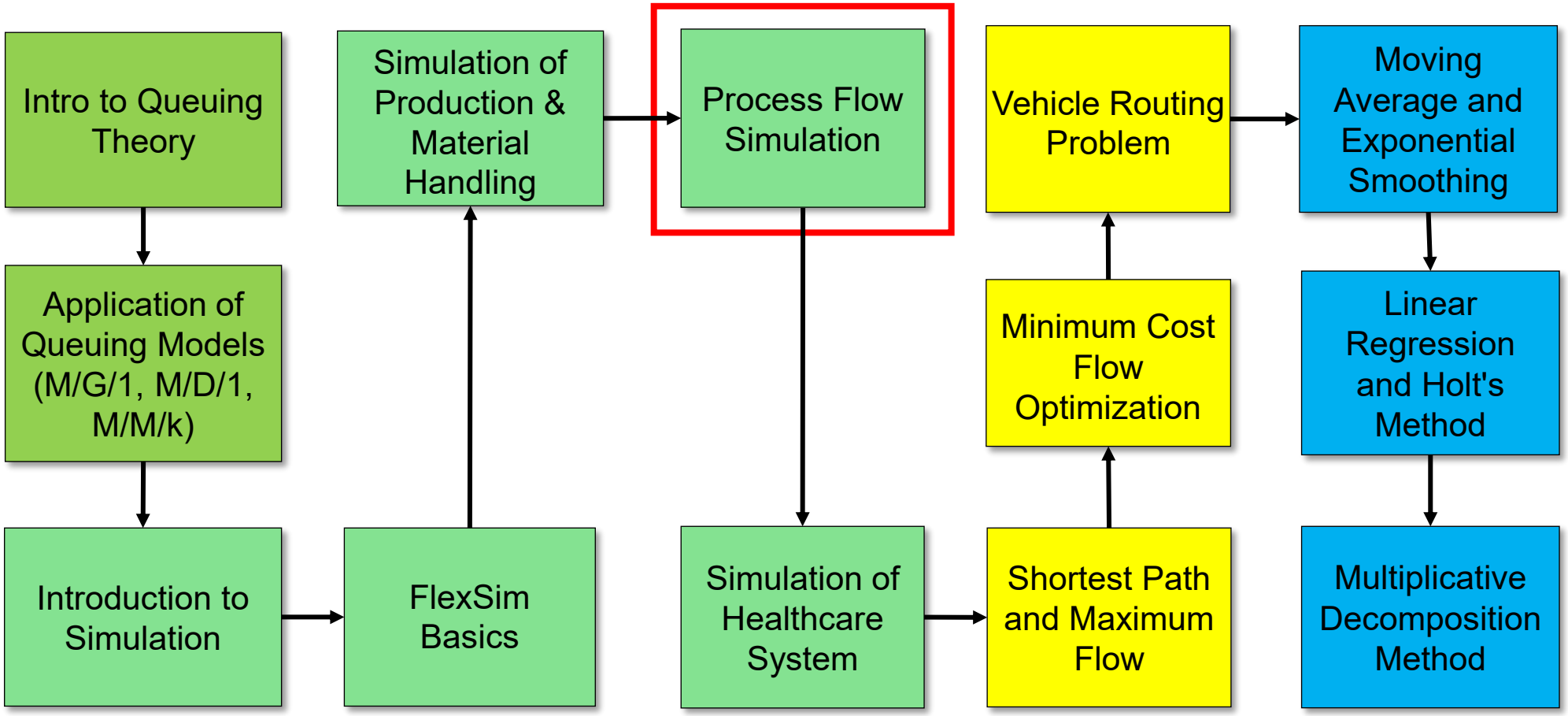
Click the **Quiz** button to edit this object

Interpret the simulation statistics of 'Order Taker' resource.

The maximum waiting time for Order Taker is _____ . The average waiting time is _____

The maximum number of people waiting in the queue is _____ . The average queue length is _____ .

The average time Order Taker performs order taking is _____ .

Average utilization of this resource is _____ . Hint: Content/Max Resource Available

E211 Progression